Parul Singh

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PROFILE SUMMARY

Generative AI Engineer with hands-on experience developing LLM-powered applications using Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, FAISS, Embeddings, and NLP. Skilled in building intelligent AI solutions with Python, Flask, Django, and modern LLM APIs, with a passion for designing scalable, production-ready Generative AI applications.

EDUCATION

BTech (CSE)- Chhatrapati Shahu Ji Maharaj University, Kanpur (CPI: 8.62/10) 2021-2025

EXPERIENCE

Indian Institute of Technology, Kanpur IITK Campus, Kanpur

Research Intern, May 2024 - July 2024

- Developed a Django-based automation system integrating multiple Electricity Regulatory Commission (ERC) sources, reducing manual document search by approximately 20 hours per week.
- Built scalable web scraping pipelines to fetch, process, and store regulatory documents from multiple government websites.
- Designed backend workflows for automated document processing, notifications, and data management.
- Collaborated with mentors and team members to deliver a production-ready research solution for real-world regulatory monitoring.

PROJECTS

The Glitch Room – AI-Powered Collaborative Learning Platform GitHub

Personal Product | July 2025 – Present

- Developed an AI-powered collaborative learning platform integrating Groq-powered LLMs for intelligent challenge generation, contextual hints, and personalized learning feedback.
- Designed prompt-driven AI workflows to evaluate user responses and generate structured performance reports highlighting strengths, improvement areas, and learning recommendations.
- Implemented Arena Challenges featuring multi-stage AI-assisted evaluation and adaptive feedback.
- Built Creator Rooms enabling collaborative challenge creation and knowledge sharing.

AI Career Coach GitHub

- Developed a Retrieval-Augmented Generation (RAG)-based AI career assistant providing resume evaluation, skill-gap analysis, and personalized career guidance.
- Integrated LangChain, FAISS, embeddings, and LLM APIs to deliver context-aware responses.
- Designed conversational AI workflows for interactive career guidance using semantic retrieval.
- Deployed the application using Flask and Render.

QueryPDF – AI-Powered Document Assistant GitHub

- Built an AI-powered document assistant enabling natural language interaction with PDF documents using Retrieval-Augmented Generation.
- Implemented semantic search using LangChain, FAISS, embeddings, and Groq-powered LLMs.
- Developed document ingestion and context-aware question-answering pipelines.
- Designed conversational AI workflows for intelligent document retrieval.

Solar Panel Defect Classification System GitHub

- Developed an EfficientNet-based deep learning model for solar panel defect classification.
- Developed a Streamlit application for real-time image upload and defect prediction.
- Implemented preprocessing pipelines and achieved accurate classification across six defect categories.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, JavaScript
- **Generative AI:** LLMs, Prompt Engineering, RAG, LangChain, FAISS, Embeddings, Semantic Search, NLP
- **Frameworks & Libraries:** Django, Flask, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Matplotlib, NLTK
- **Tools:** Git, GitHub, Render, ChatGPT, Claude, Google Gemini, Groq API
- **Databases:** MySQL

ACHIEVEMENTS

- Selected for IIT Kanpur's SURGE Program 2024 and awarded the Institute-Funded Category (stipend recipient).
- Presented "*The Glitch Room*" at Unstop Playstorm and PitchFest 2026; selected in Round 1 of Unstop Playstorm for demonstrating innovation, problem-solving, and entrepreneurial thinking.